International Islamic University Chittagong

Centre for General Education (CGED)
Midterm- Examination
Autumn -2025
Course Code: GEEM-1101
Course Title: Text of Ethics and Morality

(Faculty of science)

Marks: 30 **Duration: 1.30 hours**

Answer the following questions.

SL	Questions	Marks	CLOs	Blooms taxonomy domain
01	*Answer the following questions.* a) Who are relatives? What are there rights? d) Write down the stages of the creation of humankind.	5 X 2 =10	CLO-1	Create
	Or a) Write down the teaching Surah al-Ikhlaas b) What are the bad behaviors that hinder the piece of our society?	5 X 2 =10	CLO-1	Analyze
02	a) What is transgender? Write in details according to your lesson b) Write the different name of surah Faatiha. c) Explain the necessity of ethics and morality in human life.	4 3 3	CLO-1	Remember & Understand
03	a) What is the basic difference between Jihad and terrorism? Write in details. b) Write the lesson from Surah al-Faatiha with its meanings.	5 X 2 =10	CLO-2	Create

International Islamic University Chittagong
Center for General Education (CGED)
Mid Term Examination -Autumn 2025

Course Code: GEEL-1106
Time: 1 hours & 30 minutes

Course Title: Advanced English
Marks: 30

Reading part-15
Unseen passage

1. Read the following passage and answer the questions that follow. 01×7=07

The recognition of Palestine as a state by some major Western powers, including the UK and France, has come so late that many would argue these nations also bear some responsibilities for the thousands of preventable deaths that have occurred in Gaza. Yet, US President Donald Trump, the most powerful man on Earth, believes the recognition of a Palestinian state by key allies is effectively re warding Hamas for its attack on Israel on October 7, 2023, which started the war. According to White House Press Secretary Karoline Leavitt, "Frankly, he believes it's a reward to Hamas."

President Trump repeated the same diatribe against its allies that have independently announced recognition of Palestine at the United Nations General Assembly's 80th session. Shockingly, it's not just President Trump, who earlier this year shockingly proposed to convert Gaza into "the Riviera of the Middle East"; many leading media houses, particularly in Britain, also echoed Israeli Prime Minister Benjamin Netanyahu's outrage by likening the recognition as a surrender to terrorism. Headlines in the British media, the day after the announcement by Prime Minister Keir Starmer, are anything but an unbelievably shameful dehumanization of the Palestinian people. Picking up a few words from Hamas' reaction that the recognition by the UK, Canada and Australia were "victory for the justice of our cause," the *Daily Mail* splashed out "Hamas claims victory." *The Times* took a similar approach, though quoting families of Israeli hostages who have accused the prime minister of handing Hamas "its greatest prize." The Telegraph headline also said, "Hamas declares 'victory' after UK recognition of Palestine."

There were, however, a few exceptions who are known for their liberal and left-wing views. The *Mirror*'s front page reads "the best hope for peace," showing a photo of damaged buildings in Gaza. *The Guardian* quoted Sir Keir Starmer's claim that the decision showed "commitment to a two-state solution." But the domineering tone was of the right-wing view of anti-recognition of a Palestinian state. According to the United Nations, more media workers have been killed by Israeli forces during their ongoing military campaign in Gaza than the journalists killed in two world wars and wars in Vietnam, Iraq and Afghanistan combined. In various counts, the toll varies between 220 and 272. While covering the Gaza war, Palestinian journalists' unmatched sacrifices to tell the stories of devastation and inhumane sufferings of the innocent victims, including women and children, to the world were largely ignored by most Western media outlets. Instead, briefings by the Israeli Defense Forces' spokesperson got prominence in their coverage.

a) Who, according to the passage, should take the responsibility of deaths in Gaza?
b) *Disgraceful* – find a word from the passage that expresses the same idea?
c) Frame a sentence with "domineering tone".
d) Find a word from the passage that expresses the same meaning as "Atrocity".
e) The man with his_____ is trying to make terrors. Find a word from the passage to fill in the gap.
f) Name a media that is for the Palestinian people.
g) what do you understand by "commitment to a two-state solution"?

Seen passage

1. Read the following passage and answer the questions that follow:
Life on earth depends on the sun. Day after day we see its light and feel its warmth, but we do not often consider their origin. Yet there are many remarkable things about the sun. One is its distance from the earth. This is about ninety-three million miles. A journey of this distance, even if it could be made, would take several hundred years even in the fastest rocket. The sun is a large star. The planet earth is very small in comparison. One hundred and nine globes the size of the earth would be needed to stretch from one

side of the sun to the other. The sun makes us feel hot, even at a distance of ninety-three million miles. This is not surprising. The temperature on the sun is about ten thousand degrees Fahrenheit. But we receive only a small part of this heat. The total heat of the sun could melt a column of ice two and a quarter mile thick and ninety-three million miles high in one second. The brightness of the sun is equally astonishing. The sun gives such a bright light that 1,575 000,000,000,000,000,000 wax candles would be needed to give an equal light. This very long row of figures gives us some idea of the brilliance of the sun. As we said earlier, we receive only a very small part of the sun's heat. We also receive only a very small part of its light. This is sufficient for the growth of trees and plants, and for the existence of living creatures on earth. Too much heat and light would destroy the balance of life The heat and light from the sun come in just the right quantities for life on earth.

Answer the following questions according to the instructions given. **01×8=08**

a) Why would a journey from the earth to the sun not be possible?
b) we should make a _____ between Islam and non-Islam to understand the reality. Find a word from the passage to fill in the gap.
c) Find a synonym of the word *Surprising* from the passage.
d) Without the correct balance of heat and light, life on the earth would not be possible. True or false?
e) *Equally astonishing*- frame a sentence with the phrase.
f) Find a word from the passage that means *more than one.*
g) Explain the phrase- *brilliance of the sun.*
h) All _____ should be loyal and submissive to the almighty. Find a word from the passage to fill in the gap.

Grammar part -05

3 Answer the following questions according to the instructions given. **1×05=05**

a) He started a business to maintain his family. (Change the voice)
b) We would negotiate if he (give) attention to our proposal. Use right form of verb.
c) Mahin together with his parents were present in the seminar. (Correct the sentence if incorrect.)
d) We proposed them to discuss the issue. (Change the voice)
e) Students should make the intruders (leave) the campus. (Use right form of verb)

Writing part- 05

4. *Artificial intelligence (AI) will play a vital role to frame the life style of human beings in the coming days.* Extend this topic sentence into a paragraph 05

Or

A boy was walking alone. He was thinking of his poverty. He was obsessed with the thoughts of how he can shine in his life. Suddenly---------- complete the story.

Speaking 05

5. Speaking test will be taken by the course teacher in a convenient time. 05

International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Final Term Examination, Autumn 2025
Course Code: CSE-1121 **Course Title:** Computer Programming I
Total Marks: 30 **Time**: 90 minutes
Figures in the right hand margin indicate full marks.

Answer all of the following questions.
[XYZ are the last three digits of your student ID.
For example, if your ID is **C253176**, then **X = 1**, **Y = 7**, and **Z = 6**.]

1 (a) Write down the different types of data types in C programming. Explain what they are used for. 2

(b) Consider the following variable declarations and initial values in a C program: 3

```
int a = 5, b = 10, c = 3;
double x = 0.1, y = 0.5;
```

Show how each of the following expressions will be calculated step by step using the precedence and associativity rules.

An example has been solved for you:

a - b % c
=> a - <u>b % c</u>
=> <u>a - 1</u>
=> 4

i) a / b + x * y - (c % b)
ii) (a > b) && (x < y) || y + c

(c) Write the output of the following C code segments. If there is any error in the code, rewrite it correctly. 2

i) int distance = **X**, time = **Y**;
double speed = distance / time;
printf("%lf\n", speed);

ii) double res = **Y** % 1.5;
printf("%lf\n", res);

— OR —

What are pre-increment and post-increment operators in C? Explain with an example. Why are they used?

(d) Write a program that converts a given number of days into **months (assuming each month has 30 days)** and the remaining days. 3

Sample Input	Sample Output
265	Months = 8 Days = 25
1000	Months = 33, Days = 10
16	Months = 0, Days = 16

2 (a) Design a C program for a Train Ticket Booking System. The program will ask for the type of seat and the number of tickets, then print a receipt with the total price. 4
The ticket price for all types for adults is 400. And students get a 50% discount on the same price.

Input format: A single line containing three integers:

- Ticket type **(1 for Shovon, 2 for AC, 3 for Snigdha)**
- Number of adult tickets
- Number of student tickets

Output format: The program should display the ticket type and the total price.

Sample Input	Sample Output
2 3 2	Seat: AC Total Fare: 1600
1 2 3	Seat: Shovon Total Fare: 1400

For the first sample, Seat type = 2 (AC). And cost = 3 adults × 400 + 2 students × 200 = 1600

— OR —

Write a C program that reads a single character from the user and determines its type.

- If the character is an alphabet, the program should identify whether it is uppercase or lowercase. For uppercase letters, it should further check if the character is a vowel (A, E, I, O, U) or a consonant, and display "Uppercase Vowel" or "Uppercase Consonant" accordingly. For lowercase letters, it should perform the same check and display "Lowercase Vowel" or "Lowercase Consonant."
- If the character is a digit (0–9), the program should determine whether it is even or odd and print "Even Digit" or "Odd Digit."
- For any other character that is neither a letter nor a digit, the program should display "Special Character."

Sample Input	Sample Output
E	Uppercase Vowel
5	Odd Digit
@	Special Character

(b) Write a C program to check if a given date is valid or invalid. The program will take three integers as input: day, month, and year, respectively. 3

A date will be considered **valid** only if it satisfies all of the following conditions:

- The year is between 1700 and 2500.
- The month is between 1 and 12.
- The day is valid for the given month:
  - If the month is 2 (February), the day must be between 1 and 28.
  - For all other months, the day must be between 1 and 30.

The program should print "Valid" if all the above conditions are satisfied. Otherwise, it should print "Invalid".

Sample Input	Sample Output
30 5 2025	Valid
31 1 2020	Invalid
29 2 2024	Invalid

(c) 3

Consider the code snippet **on the right.** What will be the values of x and y when n takes the following values:
(i) 0
(ii) 5
(iii) 9
Explain your answer with rough calculations.
	int x = 4;
	int y = XY % 10;
	if (n >= 0)
	  if (n == 5)
	  x -= 4, y += 10;
	else
	  x *= 2, y -= 5;
	printf("x = %d y = %d", x, y);

3 (a) 1

Consider the code snippet of a for loop on the right, and write equivalent programs using a
i) while loop
ii) do-while loop.
	int i, sum = 0;
	for(i = 0; i < 5; i++) {
	  sum += i;
	  printf("%d %d\n", i, sum);
	}

(b) Write the output of the following code segments: 2

i)
```
int n = 2YZ;
while(n)
{
    printf("%d, ", n%10);
    n /= 10;
}
```

ii)
```
int i = 1;
while (i <= 100)
{
    if (i * i > XY)
        break;
    printf("%d\n", i);
    i++;
}
```

(c) You are given two C programs that find the square root of a given number n. Analyze the programs and determine which one is more efficient and finishes earlier. Explain your answer. 3

Code A	Code B

`int n = 1024;`
`int i = 0;`
`while (i * i <= n) {`
`  i++;`
`}`
`printf("Square root of %d ≈ %d\n", n, i - 1);`
	`int n = 1024;`
	`int low = 0, high = n, mid;`
	
	`while ((high - low) > 1) {`
	`  mid = (low + high) / 2;`
	`  if (mid * mid > n)`
	`  high = mid;`
	`  else`
	`  low = mid;`
	`}`
	`printf("Square root of %d ≈ %d\n", n, low);`

(d) You are given an initial salary. Each year, the salary increases by 5% of the current salary. Write a C program to find how many years are required for the salary to become at least 2000. 4

Sample Input	Sample Output
1000	15
1500	6

For the first sample, initial salary = 1000

- After 1st year = 1000 + 5% of 1000 = 1050.00
- After 2nd year = 1050 + 5% of 1050 = 1102.50

...

- After 15th year = 1979.93 + 5% = 2078.93

— OR —

Write a C program to check whether a given integer is an Armstrong number.
[An Armstrong number is a number that is equal to the sum of its own digits, each raised to the power of the **number of digits.**

Example: 153 is an Armstrong number because $1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$.]

Sample Input	Sample Output
9474	Armstrong number
123	Not an Armstrong Number

International Islamic University Chittagong
Department of Computer Science and Engineering

Mid-term Examination, Autumn-2025 | **Program: B.Sc. Engineering in CSE**

Course Code: **EEE-1121** | Course Title: **Basic Electrical Engineering**

Time: 1 hour 30 minutes | Full Marks: 30

[Answer all the questions from the following. Figures in the right margin indicate full marks]

1) a) Differentiate between active and passive elements with examples. Find the v_o and i from **Fig.1a** CO-1 U 05

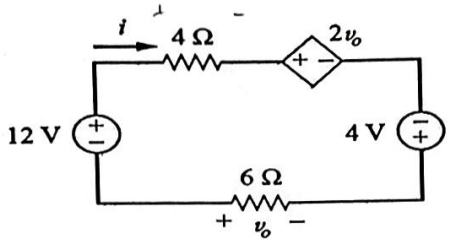

Fig- 1a

1) b) Discuss the concept of current divider rule and find the value of i_1 to i_5 in the circuit of **Fig.1b** CO-2 Ap 05

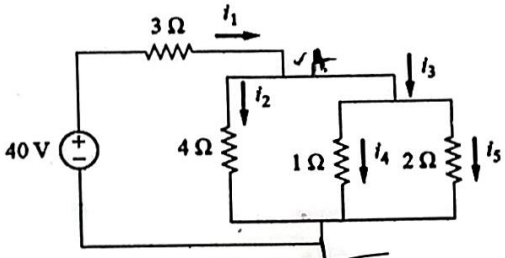

Fig- 1b

2) a) Define node and branch in an electrical network. Find the node voltages of the network of **Fig. 2a** CO-1 U 05

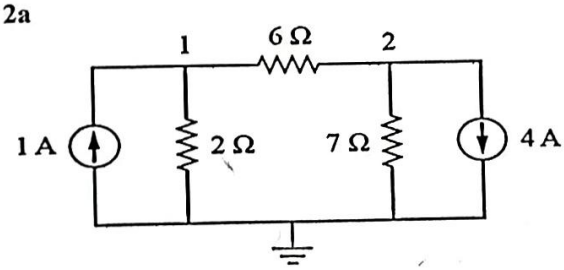

Fig- 2a

2) b) For the circuit in **Fig. 2b**, find the branch currents i_1 and i_2 using mesh analysis. CO-2 An 05

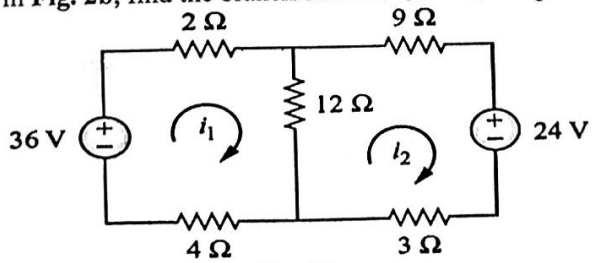

Fig: 2b

3) a) State superposition theorem and using superposition theorem find v_o in the circuit of **Fig. 3a** CO-2 Ap 05

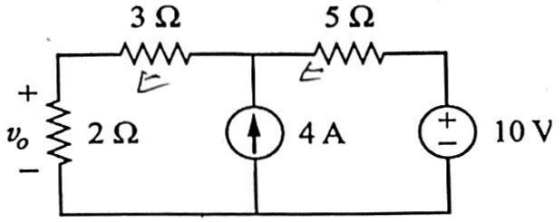

Fig- 3a

3) b) Using Thevenin's theorem, find the Thevenin equivalent circuit to the a-b terminal of **Fig.3b**. CO-2 E 05

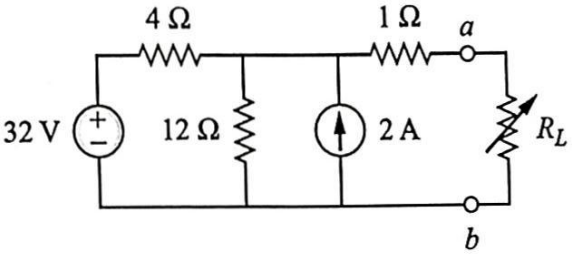

Fig- 3b

OR,

3) a) How to convert a voltage source to a current source & current source to a voltage source? Use source transformation to find *i* in the circuit of **Fig. 3c** CO-2 Ap 05

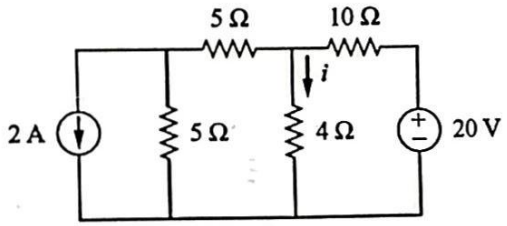

Fig- 3c

3) b) Find the value of R_L for maximum power transfer in Fig. 3d and also find the maximum power. CO-2 E 05

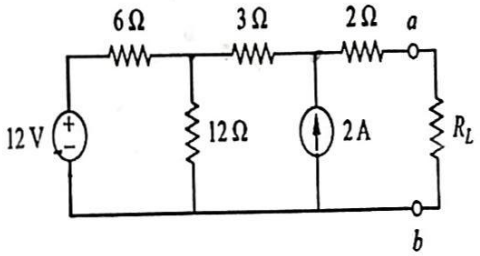

Fig-3d

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Mid Term Examination

Program: B. Sc. in CSE
Course Code: MATH-1107
Time: 1:30 hours

Semester: Autumn 2025
Course Title: Mathematics-I
Total Marks: 30

(i)	Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii)	Please answer the several parts of a question sequentially.
(iii)	Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1:	Compute the functions, limit and continuity of a function, derivatives, integrals and extrema of single-variable and/or multivariable functions.
CLO2:	Understand the techniques of differentiation and integration.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

		Marks	CLO	DL
1. a)	Define Domain and Range of a function.	1	CLO1	R
b)	Find the domain and range of the function, $f(x) = \sqrt{x^2 - 9}$	2	CLO1	U
c)	Test the continuity of the following function at $x = 0$ $$y = f(x) = \begin{cases} \frac{1-\cos x}{x^2}; & x \neq 0 \\ 1; & x = 0 \end{cases}$$	4	CLO1	U
Or.	Test the differentiability of the following function at $x = 1$ $$y = f(x) = \begin{cases} 5x - 4; & 0 < x \leq 1 \\ 4x^2 - 3x; & 1 < x < 2 \end{cases}$$			
d)	Evaluate the limit using L'Hospitals rules $\lim_{x \to \infty} (e^x + 1)^{\frac{-2}{x}}$	3	CLO1	U

2.	a)	Find the differential co-efficient of $\ln x$ by the geometrical method.	3	CLO2	U
	b)	Differentiate $\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$ with respect to $\sec^{-1}\frac{1}{2x^2-1}$	4	CLO2	U
	c)	If $y = (cosx)^{(cosx)^{(cosx)^{\cdots\infty}}}$ then find $\frac{dy}{dx}$	3	CLO2	U
3.	a)	If $x = \sin\left(\frac{1}{m}\ln y\right)$ then using the Leibnitz theorem show that, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$	4	CLO2	U
	b)	Verify the Roll's theorem for the function $f(x) = x^2 - 3x + 2$ in the interval (1,2)	3	CLO2	U
	Or.	,Obtain the Maclaurin's series generated by the function $f(x) = \ln(1+x)$			
	c)	Verify Mean Value theorem for the function $f(x) = \frac{x^3}{4} + 1$ over the interval [0, 2].	3	CLO2	U
	Or.	Expand $\frac{1}{x}$ in ascending power of $(x-2)$ by Taylor's Series.			

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE
Mid-Term Examination, Autumn-2025
Course Code: **PHY-1101** Course Title: **Physics-I**
Time: 1 hour 30 minutes Full Marks: 30

(i) Answer **all** the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes and Bloom's Levels are mentioned in additional columns.

1.	a)	Define the center of mass of a system.	CLO1	R	1
	b)	State and prove the parallel axis theorem for the moment of inertia.	CLO2	U	6
	c)	A solid disc flywheel of mass 20 kg and radius 30 cm is rotating at 100 rad/s. Calculate its rotational kinetic energy.	CLO3	A	3
2.	a)	Derive an expression for the gravitational potential at an external point of a uniform spherical shell .	CLO2	U	7
	b)	A, thin spherical shell of mass $YY\times10^{24}$ kg and radius 5000 km is given. A test mass of Z kg is placed at a distance of 4000 km from the center. Calculate the gravitational potential at this point. Here, YY is the last two digits of your ID and Z is the mass of your body.	CLO3	A	3
3.	a)	Define a wavefunction in quantum mechanics.	CLO1	R	1
	b)	Derive an expression for the time-independent Schrödinger wave equation.	CLO2	U	6
		Or			
		Derive an expression for the gravitational potential at a point due to a point mass, and hence calculate the gravitational field intensity at that place.			
	c)	Find the uncertainty in the momentum of a particle when its position is determined within 0.01 cm. Find also the uncertainty in the velocity of an electron and an alpha particle respectively, when they are located within 5×10^{-8} cm.	CLO3	A	3
		Or			
		If a star of mass 2×10^{31} kg collapses under its gravity, using the escape velocity formula, calculate the radius at which escape velocity equals the speed of light.			

International Islamic University Chittagong

Department of Computer Science & Engineering

Mid Term Lab Examination, Autumn 2025

Sec: 1BM

CSE 1122 Computer Programming with C Lab

Total marks: **30** Time: **60** minutes

[Answer all of the following questions. Figures in the right-hand margin indicate full marks.]

1.	An electric power distribution company charges its domestic consumers as follows:
	
	Consumption Units — *Rate of Charge*
	0 - 200 — Rs. 0.50 per unit
	201 - 400 — Rs. 100 plus Rs.0.65 per unit excess of 200
	401 - 600 — Rs. 230 plus Rs.0.80 per unit excess of 400
	601 and above — Rs. 390 plus Rs.1.00 per unit excess of 600
	
	Now write a program that reads the customer number and power consumed and prints the amount to be paid by the customer.
	
	`Enter CUSTOMER NO. and UNITS consumed 202  225`
	`Customer No:202 Charges = 116.25`
		CLO 3	12

2.	Write a program to calculate the CGPA Grade of a student based on marks. Marks Range — Grade 80-100 — A+ 4 70-79 — A 3.75 60-69 — A- 3.5 50-59 — B+ 3.25 40-49 — B- 3.00 <40 — F 0 There are n subjects. Take n as input followed by the marks of n subjects. **Sample1:** Enter marks of 5 subjects: 85 74 69 58 90 Your CGPA is: 3.50 **Sample 2:** Enter marks of 5 subjects: 85 49 70 68 60 Student has failed in one or more subjects. Your CGPA is: 2.89	CLO2	10
3	Write a C program that takes a positive integer N from the user and uses a for loop to calculate and display the sum of the first N natural number. Sum = 1+2+3+4+…+N **Sample I/O:** Enter a positive integer: 10 Sum of first 10 natural numbers I : 55	CLO1	8